

Engine Evaluation Tool - Technical Report

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1.1 Project General Description

This calculation tool was developed in Python with the aim of automating the preliminary evaluation of an internal combustion engine. This code takes a dedicated Excel template file as input and, using internal formulas and an object (Dataclasses) oriented approach, provides an estimation of the engine performance and a verification of the critical sections of its components.

1.2 Code and Modules architecture

The software architecture is divided into the following sections to ensure a better scalability and easy maintenance in the future:

Module	Functional Description
InputData.py	Input data management
PreliminaryCalculation.py	Main engine data calculation including mean engine speed and valve diameter
Engine_Calculation_Tool.py	Main script that manages modules, calculation and report generation
Thermodynamic.py	Kinematic and Thermodynamic calculation
Pressure_scaling.py	Scaling of the pressure curve as a function of the new PFP
Material.py	Computes some useful data for the subsequent structural analysis
ValveDynamic_Cynimatic.py	Kinematics and Dynamics of the intake valves
Engine_force.py	Kinematic and Dynamic calculation of the piston motion
CrankLoad.py	Summary of the main loads acting on the engine components
PistonCaclulation.py	Structural analysis of the piston
PistonRing.py	Structural analysis of the piston ring
PistonPinCalculation.py	Structural analysis of the piston pin
Plot.py	Automatic generation of graphs
Report.py	Automatic generation of a technical report

1.3 Notes

This tool requires the following Python lybraries: Dataclasses, Numpy, Pandas, Matplotlib and ReportLab

2.1 Base Engine Data

The Base Engine Data are reported in the following table:

Parameter	Value	Unit
Engine Displacement	600.0	cm3
Engine Cycle	4	-
Number of Cylinders	4	-
Piston Stroke	42.5	mm
Piston Bore	67.0	mm
Engine Speed @ Maximum Power	13500	RPM

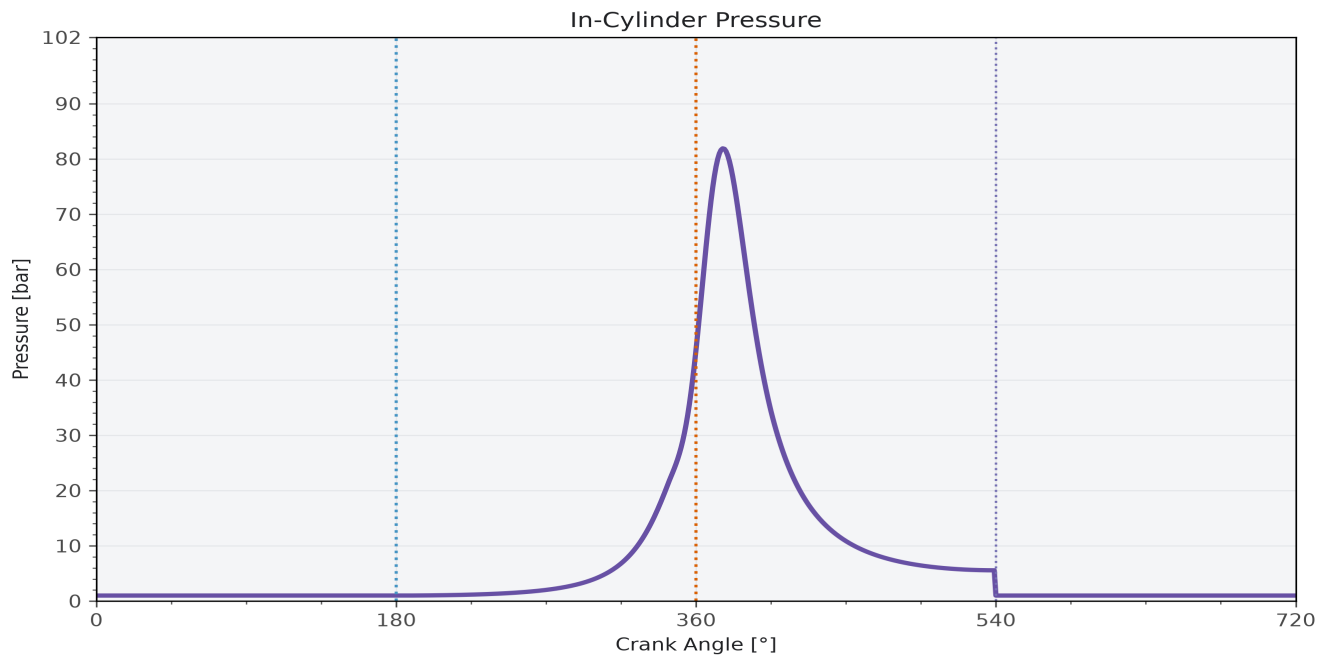
2.2 Additional Engine Data

The following data are needed to further define the engine:

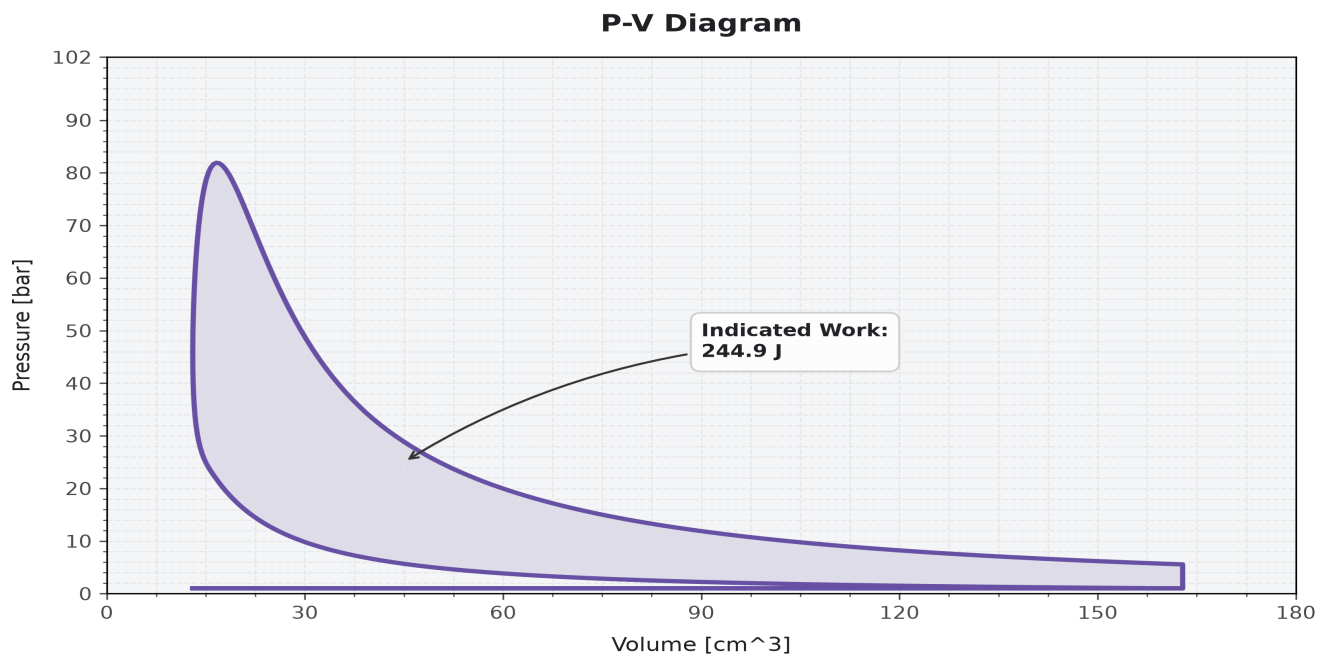
Parameter	Value	Unit
Mean Piston Speed	19.1	m/s
Number of Intake Valves	2	-
Number of Exhaust Valves	2	-
Intake Valve Diameter	25.0	mm
Exhaust Valve Diameter	17.5	mm

2.2 Engine Thermodynamic Data

The in-cylinder pressure computed by the Wiebe function is the following:



The indicated cycle and useful work is depicted in the following picture:

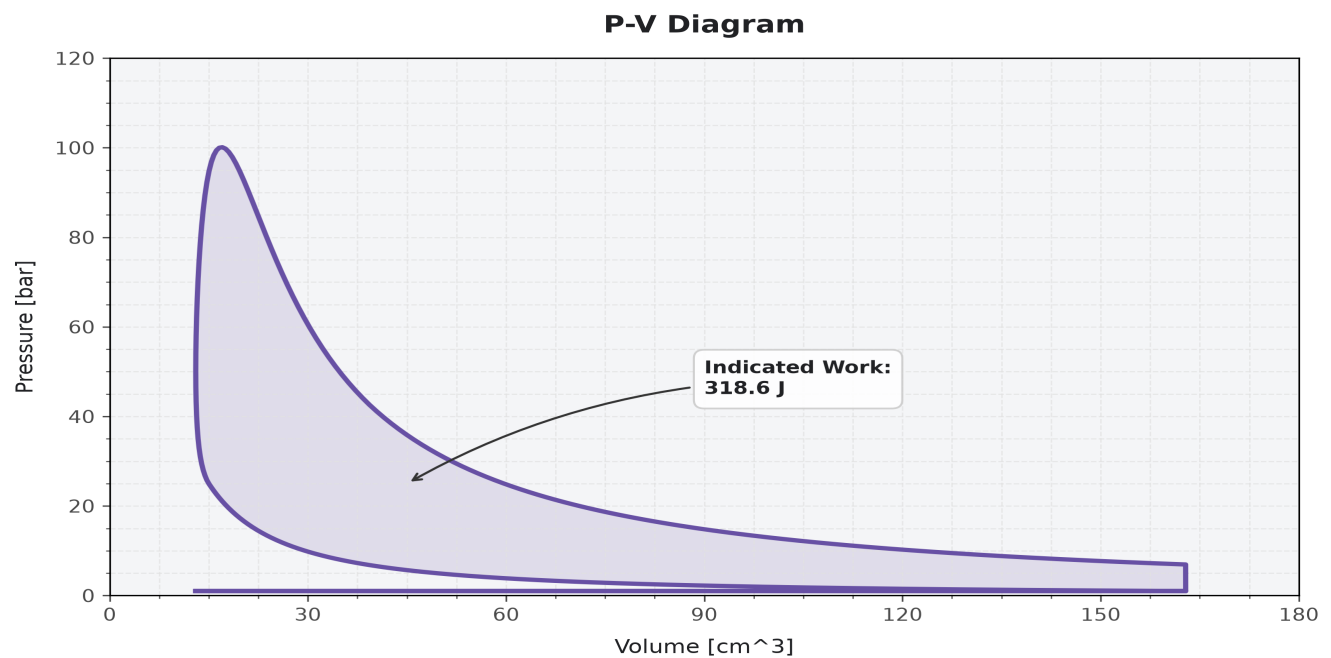
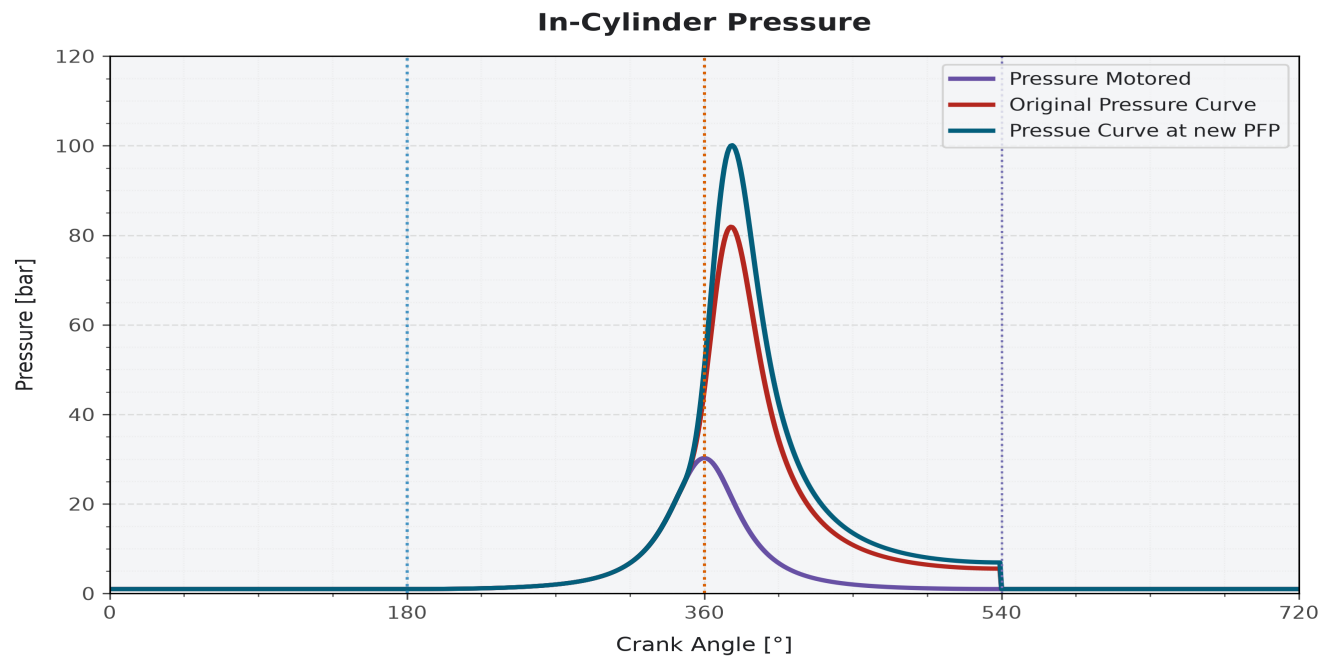


The table reports the general engine performance data:

Parameter	Value	Unit
Cylinder Theorical Air Mass	0.2	g
Cylinder Fuel Mass	0.014	g
Cylinder Heat	616.1	J
Engine Power	277.3	KW
Engine Power	124.4	CV

3.1 Engine Performance at New Peak Firing Pressure

The new in-cylinder pressure computed by the Wiebe function and the new indicated work are depicted in the following pictures:



4. Valve Dynamic and Kinematic

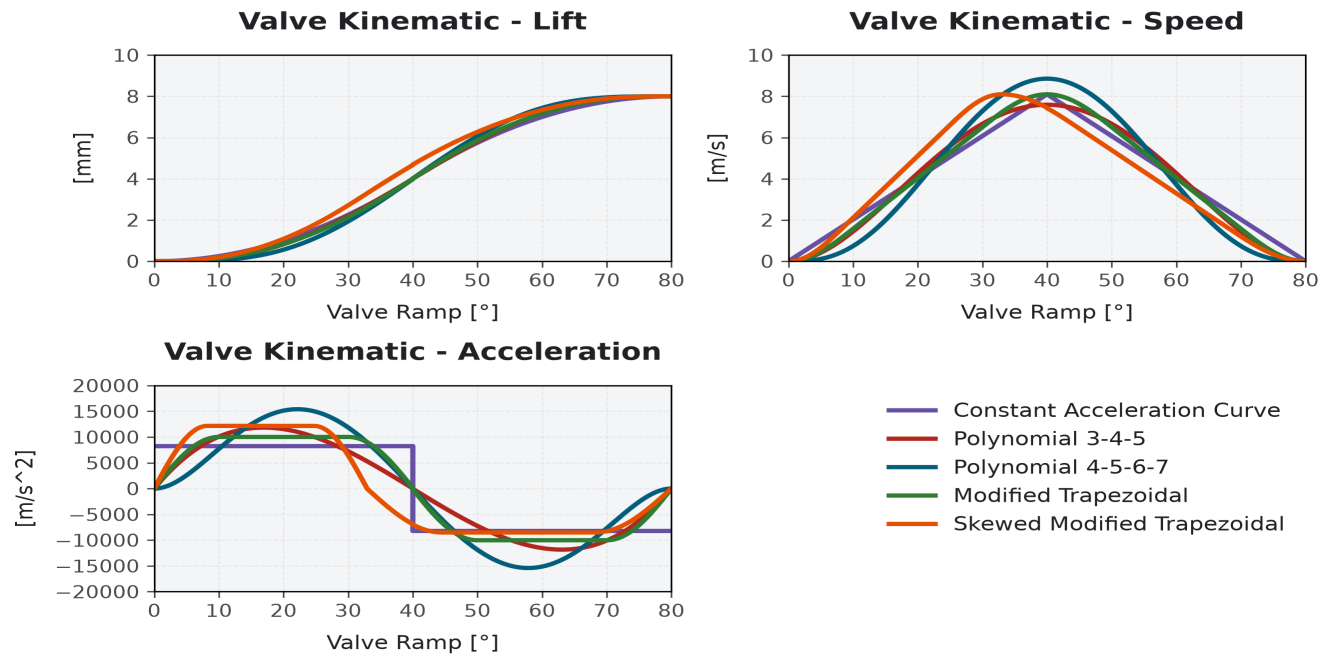
In the following chapter the calculation about the valve kinematics and dynamics are reported.

In addition to that the following useful information are listed:

- 1) All the formulas use a DWELL boundary condition with the exception of the constant acceleration curve;
- 2) The Constant Acceleration Curve is reported as a reference. Although it produces the overall better balance between the positive and negative acceleration, the jerk is too high;
- 3) This computation refers only to the intake valve; the number of intake valves in this engine are: 2

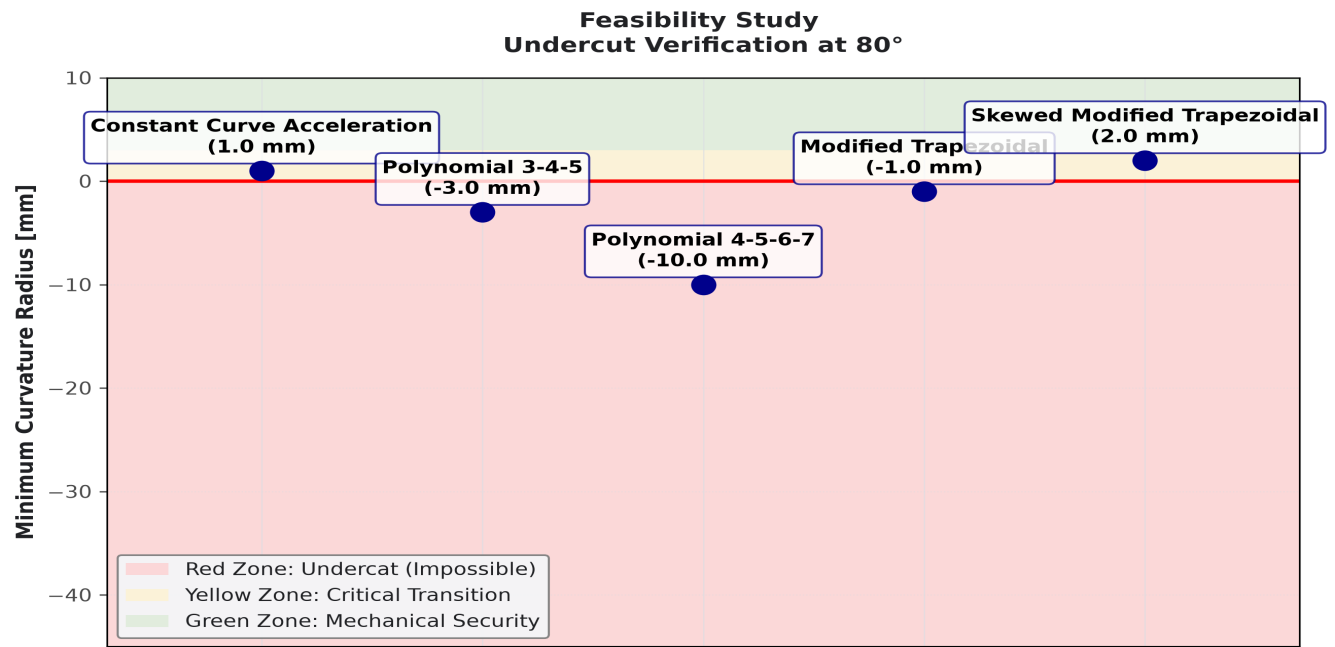
4.1 Valve Lift, Speed and Acceleration

The lift, speed and acceleration profiles of intake valve are depicted:



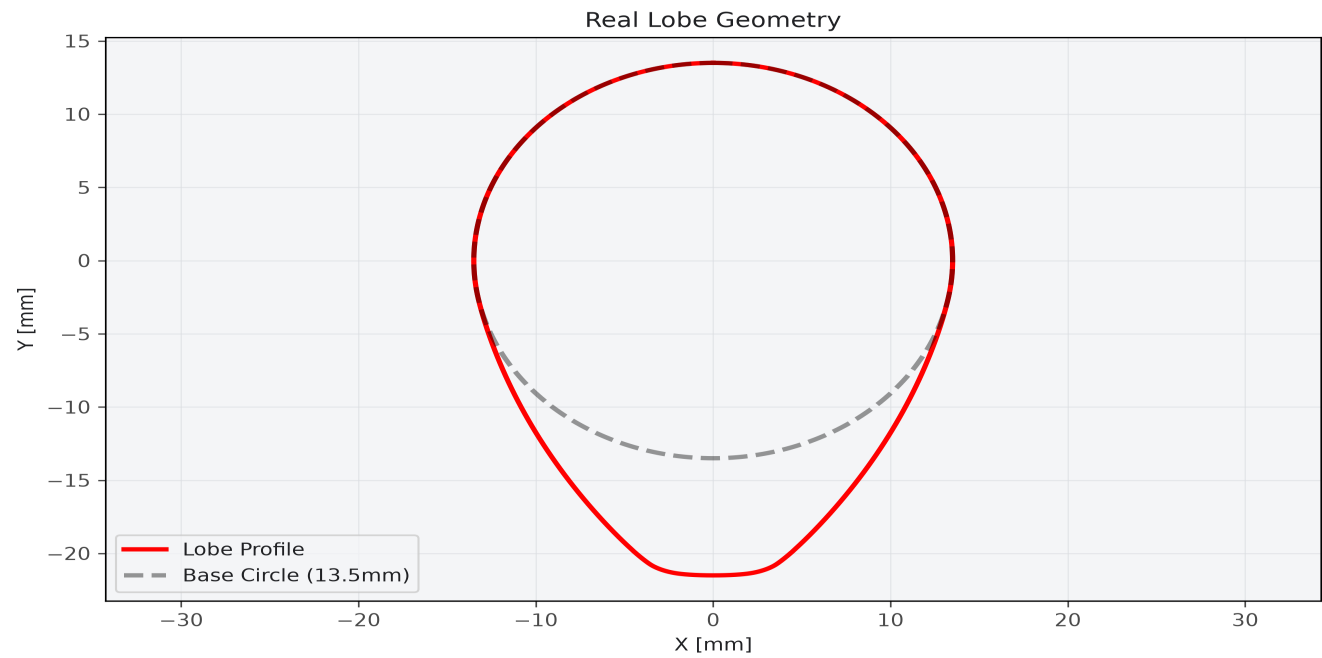
4.2 Feasibility Study

This is a check to understand which profile it is possible to realize:



4.3 Lobe Profile

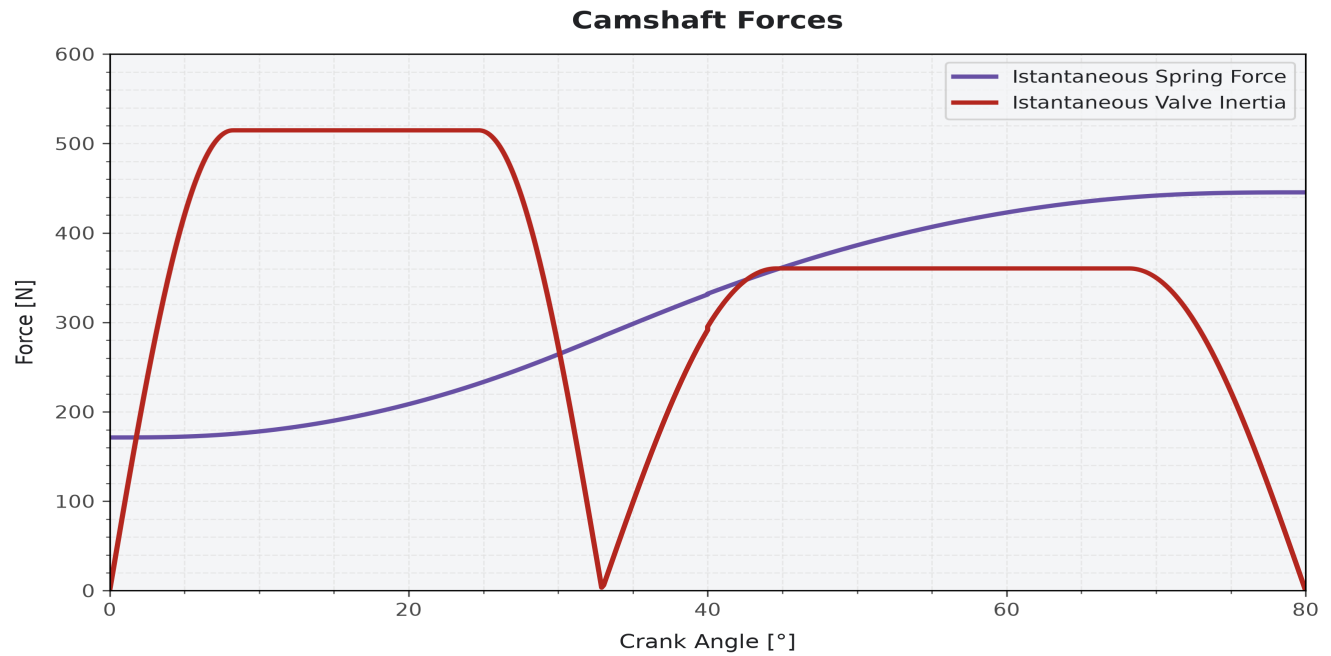
This is the lobe profile:



4.4 Spring Dimensioning

The valve Spring is calculated using the following input data:

- 1) Material: VD CrV
- 2) Preload: 5mm



The table reports the data about the spring dimensioning:

Parameter	Value	Unit
Number fo Coil	4	-
Maximum Torsional Stress	1097.2	MPa
Safety Factor	0.6	-
Spring Computed Stiffness	34.3	Nmm
Spring Effective Stiffness	31.0	Nmm

4.5 Spring Frequency

The table reports the data about the lobe and spring frequency at: 13500RPM

Parameter	Value	Unit
Camshaft Frequency	112.5	Hz
1° Spring Fundamental Frequency	477.1	Hz
Frequency Ratio	4.2	-

4.6 Spring Calculation Remark

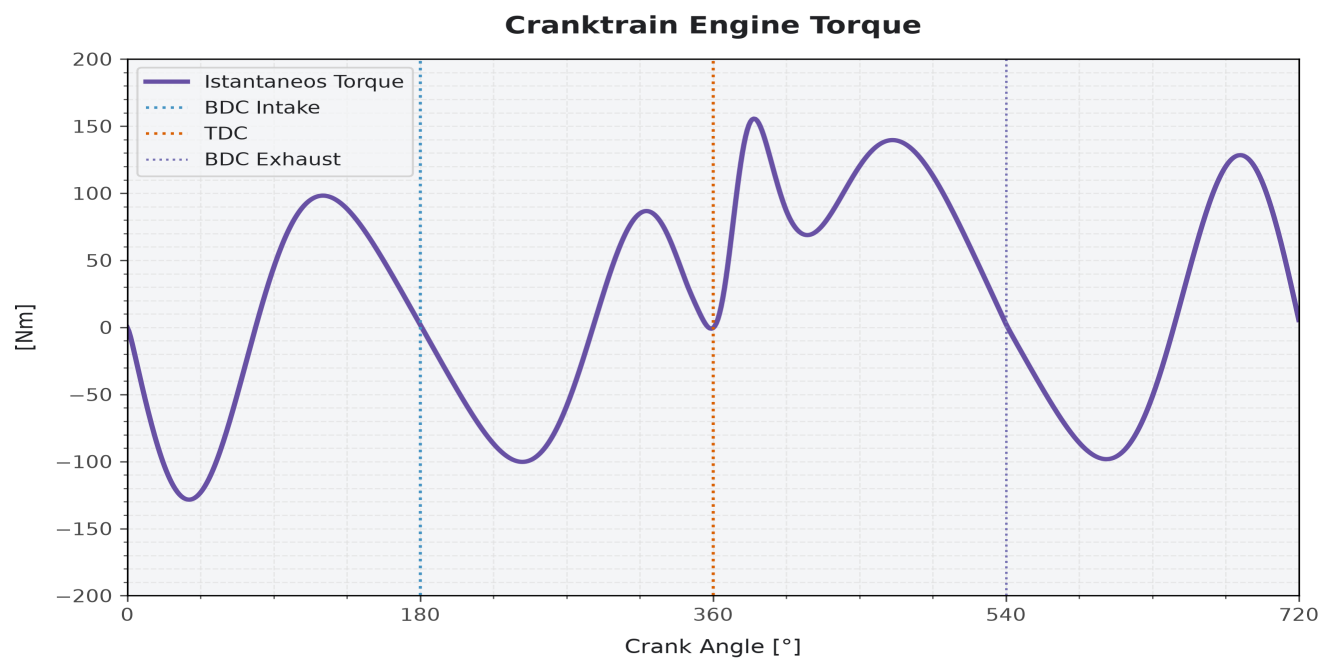
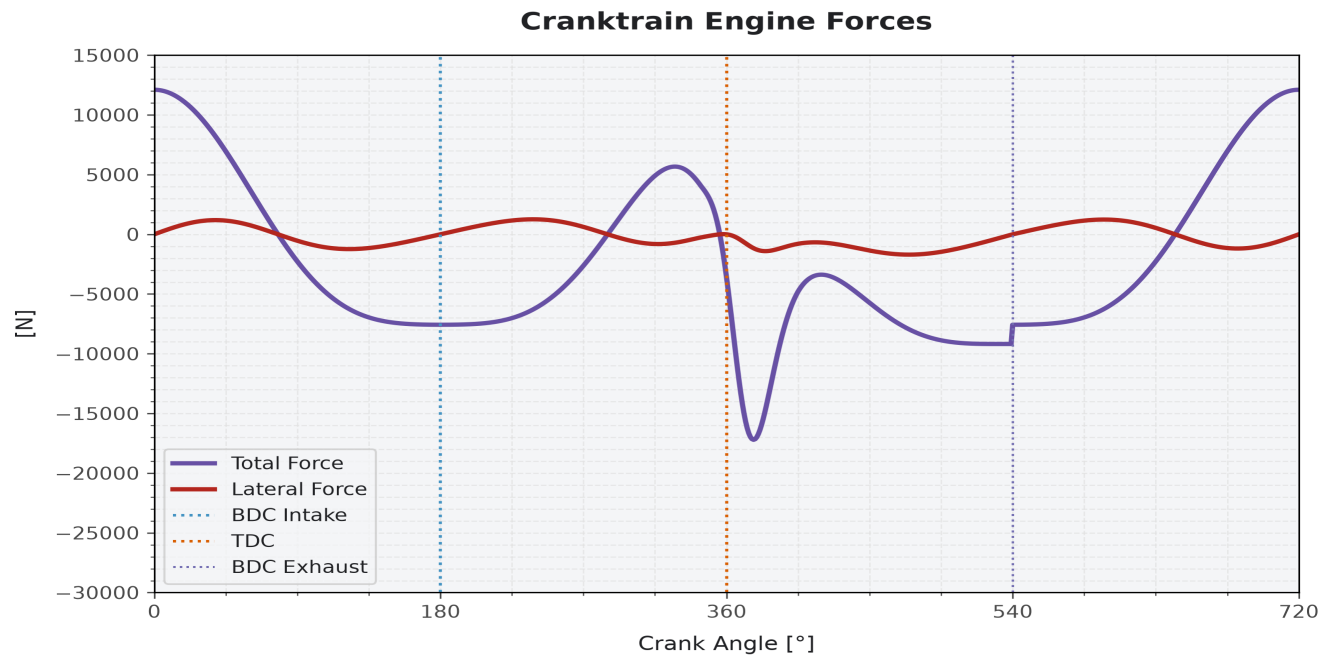
The data show that this material or configuration is not suitable for this application. It is suggested to change the material or move towards a dual parallel spring configuration

5.1 Cranktrain Engine Forces and Torque

The following two forces are computed:

- 1) The total force due to the components' inertia and gas force;
- 2) The instantaneous torque that is produced by the engine.

Be aware that the following data refer to a single engine cylinder.



5.2 Cranktrain Engine Forces

In this chapter there is a table that resumes the forces acting on the cranktrain that will be used for the validation of the components critical sections

Parameter	Value	Unit
Maximum Compression due to Gas Force	-28526.3	N
Alternating Mass Inertia Force at TDC	8894.5	N
Alternating Mass Inertia Force at BDC	-5564.0	N
Connecting Rod Inertia Force at TDC	3211.8	N
Connecting Rod Inertia Force at BDC	-2009.1	N
Connecting Rod Rotating Inertia	10057.6	N

6.1 Piston Base Dimensions

The table report the data relative to the piston main dimensions

Parameter	Value	Unit
Piston Bore Diameter	67.0	mm
Piston Total Height	73.7	mm
Piston Skirt Height	46.9	mm
Piston Crown Wall Thickness	6.0	mm
Piston Crown Thickness	6.0	mm
Number of Oil Holes	10	-
Oil Holes Diameter	1.2	mm
Piston Ring Land Thickness	2.7	mm
Compression Ring Radial Thickness	3.0	mm
Oil Ring Radial Thickness	3.0	mm
Piston Ring Width	3.0	mm

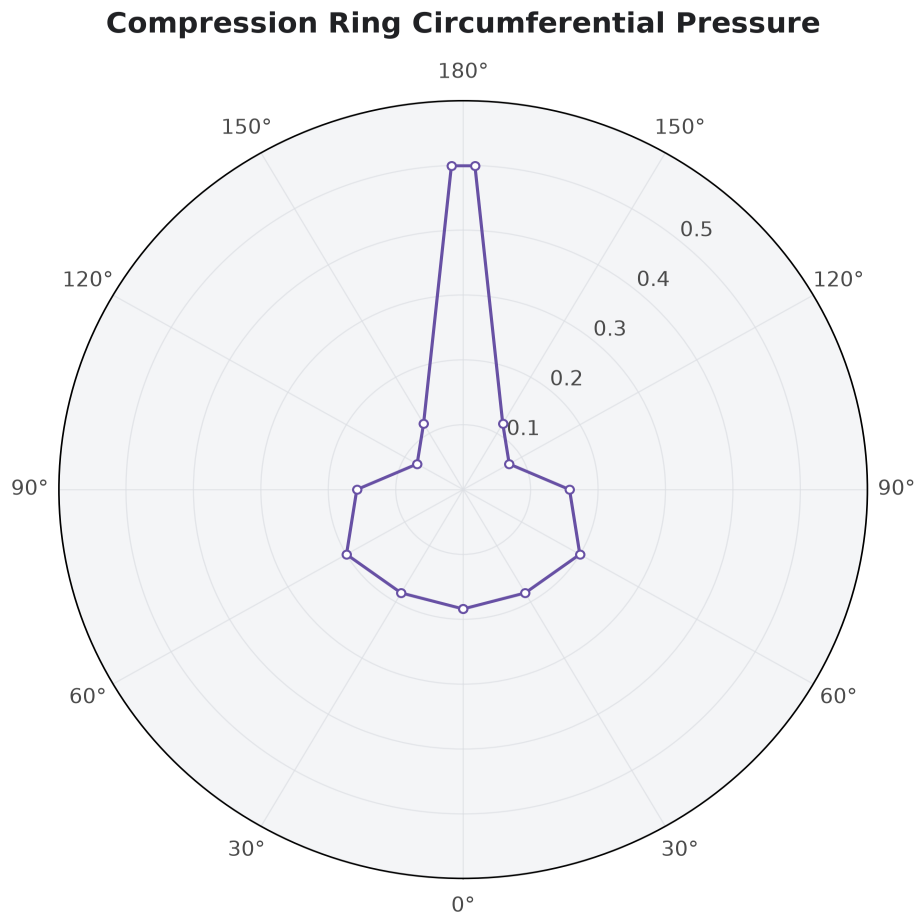
6.1 Piston Verification

The table reports values for the usual stresses exerted by the piston

Parameter	Value	Reference	Unit
Crown Bending Stress	126.7	50 - 150	MPa
Crown Heat Stress	46.3	-	MPa
Crown Total Stress	173.1	40 - 75	MPa
Crown Compression Stress	13.7	30 - 40	MPa
Crown Rapture Stress	1.7	4 - 10	MPa
Ring Land Shear Stress	6.4	-	MPa
Ring Land Bending Stress	23.0	-	MPa
Ring Land Combinated Stress	26.4	30 - 40	MPa
Maximum specific pressure exerted by the piston skirt	0.4	0.3 - 1	MPa
Maximum specific pressure exerted by the entire piston height	0.3	0.2 - 0.7	MPa

7.1 Compression Ring

The following picture depict the circumferential pressure [MPa] for a cast iron compression ring



The following table resumes some useful data:

Parameter	Value	Reference	Unit
Average Radial Pressure Wall	0.2	0.11 - 0.37	MPa
Ring Bending Stress in Operating State	208.2	-	MPa
Ring Bending Stress when Slipping over a Piston	389.6	220 - 450	MPa

7.2 Compression Ring Remark

It is suggest to check the material or goemetry characteristics of the compression ring. It could not be suitable for this application.

8.1 Piston Pin Geometry

The following table report the piston pin overall geometry data:

Parameter	Value	Unit
Piston Pin Lenght	45.5	mm
Piston Pin External Diameter	14.0	mm
Piston Pin Thickness	3.0	mm
Piston Pin Internal Diameter	8.0	mm
Piston Pin Cross-sectional Area	103.7	mm ²

8.2 Piston Pin - Middle Section Validation

The following tables resume the verification results for the pin middle section:

Parameter	Value	Unit
Type Load	Start-up condition	-
Bending Stress	674.1	MPa
Ovalizing Stress on the Outer Fiber	-241.7	MPa
Equivalent Stress	822.1	MPa
Safety Factor	1.5	-

Parameter	Value	Unit
Type Load	Overlap Condition	-
Bending Stress	210.2	MPa
Ovalizing Stress on the Outer Fiber	-75.4	MPa
Type Load	Combustion Phase	-
Bending Stress	463.9	MPa
Ovalizing Stress on the Outer Fiber	-166.4	MPa
Safety Factor	1.2	-

8.3 Piston Pin - Shear Cross-Section Validation

The following tables resume the verification results for the shear cross-section:

Parameter	Value	Unit
Type Load	Start-up condition	-
Average Tangential Stress	137.6	MPa
Maximum Tangential Stress	328.9	MPa
Ovalizing Stress Inner Fiber	-351.2	MPa
Normal Stress	-104.5	MPa
Ideal Sigma	729.5	MPa
Safety Factor	1.6	-

Parameter	Value	Unit
Type Load	Overlap Condition	-
Average Tangential Stress	42.9	MPa
Maximum Tangential Stress	102.5	MPa
Ovalizing Stress Inner Fiber	-109.5	MPa
Normal Stress	-32.6	MPa
Type Load	Combustion Phase	-
Average Tangential Stress	94.7	MPa
Maximum Tangential Stress	226.3	MPa
Ovalizing Stress Inner Fiber	-241.7	MPa
Normal Stress	-71.9	MPa
Safety Factor	1.1	-